

Vertebral osteomyelitis and a forgotten dog bite in an immunocompetent patient

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Abstract

A woman in her 60s presented with a three-week history of progressive subacute back pain without neurological deficits. MRI of the lumbar spine demonstrated necrosis at the L3–L4 disc and adjacent vertebral bodies, raising suspicion for malignancy or infection. Two image-guided bone biopsies were nondiagnostic. A peripheral infectious disease workup was similarly unremarkable. Owing to the persistent concern for infection, 16S ribosomal RNA sequencing was performed on biopsy samples, which identified *Capnocytophaga canimorsus*. Only after receiving this result did the patient recall a minor dog bite sustained six weeks prior to symptom onset. She was managed with an extended course of oral antibiotics with complete clinical recovery. This case illustrates the capacity of *C. canimorsus* to cause insidious vertebral osteomyelitis in immunocompetent individuals, underscores the diagnostic utility of 16S rRNA sequencing in culture-negative infections, and highlights the critical importance of a detailed exposure history in atypical clinical presentations.

Keywords: vertebral osteomyelitis; *Capnocytophaga canimorsus*; immunocompetent

Introduction

C. canimorsus is a slow-growing Gram-negative anaerobic bacterium commonly found in the oral mucosa of cats and dogs. It is a zoonotic pathogen transmitted to humans through bites or scratches, leading to infections at various body sites, including vertebral osteomyelitis [1–3]. Although such infections are most frequently observed in immunocompromised patients, they may also occur in immunocompetent hosts [4]. Due to its fastidious growth in microbiological media, polymerase chain reaction (PCR) is considered the gold standard diagnostic tool for detecting *C. canimorsus* infections [5]. *Capnocytophaga* species are typically sensitive to penicillins, third-generation cephalosporins, carbapenems, clindamycin, chloramphenicol, and doxycycline. Post-exposure prophylaxis is recommended in immunocompromised patients [5].

Case report

A woman in her early 60s was referred by her general practitioner with a three-week history of progressive back pain, without focal neurological symptoms. She denied fever, infective symptoms, or trauma, and reported no weight loss, anorexia, or B symptoms. She later recalled a minor dog bite sustained six weeks prior, while intervening in a dog attack, which healed quickly and was initially disregarded. Her past history was unremarkable except for remote melanoma excision, and there was no other history of immuno-

suppressive conditions. Clinical examination revealed focal lower lumbar spine tenderness without neurological deficits.

On admission, her white cell count was $12.1 \times 10^9/L$ with neutrophils $9.17 \times 10^9/L$, but subsequently normalized. Her CRP was 51 mg/L (<5 mg/L), ESR 45 mm/hr (<20 mm/hr), and procalcitonin 0.26 µg/L (0.5–2.0). Other haematological, biochemical, oncological, and rheumatological evaluations were normal. An initial community MRI (Fig. 1A) demonstrated abnormal marrow signal intensity involving the L4 vertebral body and pedicle, the L3/L4 intervertebral disc, and the inferior endplate of L3, without aggressive bone destruction. Repeat MRI performed two weeks later (Fig. 1B) showed enhancement and bone marrow signal loss within L3 and L4 vertebral bodies showing definite progression consistent with discitis-osteomyelitis without abscess formation. A PET scan, undertaken due to melanoma history, revealed progressive bone erosion and a new right paraspinal intensely avid soft tissue lesion suggestive of a phlegmon (Fig. 2).

A CT guided percutaneous needle biopsy, performed two days after the repeat MRI, demonstrated only cartilage tissue, consistent with sampling error. A repeat CT guided biopsy two weeks later showed suppurative inflammatory exudate; however, microscopy and cultures, including for acid-fast bacilli, were negative. Given the high suspicion of infection, stored second biopsy specimen was subjected to 16S ribosomal RNA testing at the Royal Brisbane and Women's hospital laboratory, which identified *C. canimorsus*.

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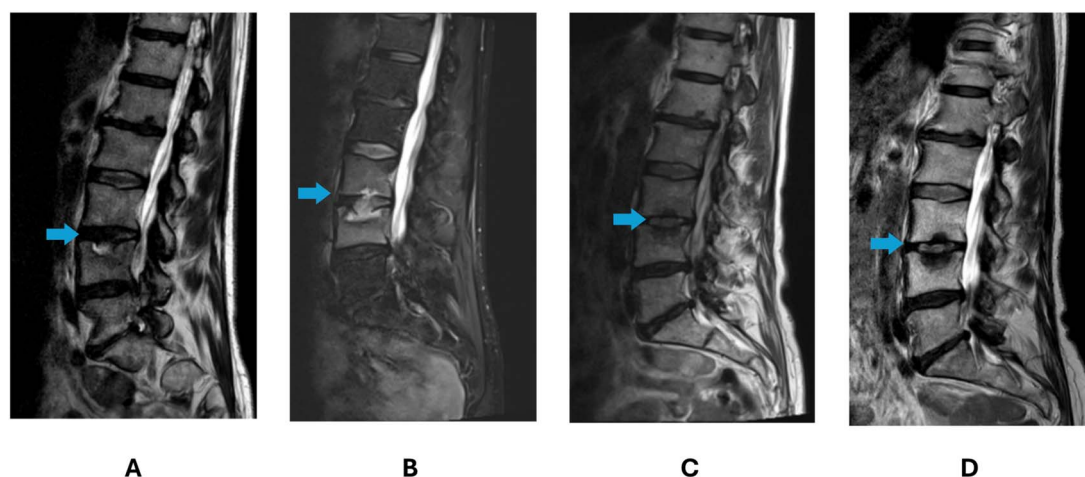


Figure 1. MRI Lumbo-sacral spine—T2 sequences. (A). Initial MRI. (B) Progression of lesion 2 weeks post initial MRI. (C) At the end of 3 months of antibiotic therapy. (D) 3 months post completion of antibiotic therapy.

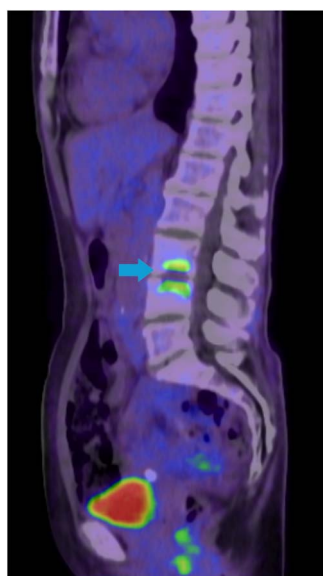


Figure 2. PET scan.

At presentation, she received a single dose of intravenous flucloxacillin 2 g, but further therapy was withheld on infectious diseases advice until biopsy. She was briefly commenced on oral Augmentin Duo Forte following the initial biopsy, which was discontinued due to lack of evidence for infection. After *C. canimorsus* was identified, she was commenced on oral ciprofloxacin 750 mg twice daily for 12 weeks. However, within one week she developed severe elbow and knee tendonitis necessitating cessation. She was then switched to oral Augmentin Duo Forte 875 mg/125 mg twice daily for a total duration of 12 weeks, which she completed successfully.

At completion of therapy, she reported significant improvement in back pain, with only mild residual positional discomfort. Follow-up MRI (Fig. 1C) showed residual changes at L3–L4 with endplate irregularity and marrow oedema. Three months later, MRI at her local hospital (Fig. 1D) demonstrated persistent but improving changes consistent with osteomyelitis/discitis with partial healing. Given her clinical recovery, ongoing observation was recommended without further imaging.

Discussion

Capnocytophaga species are recognized to cause local and systemic infections predominantly in immunocompromised patients, though cases in immunocompetent individuals have also been reported. There are previous reports of vertebral osteomyelitis caused by *Capnocytophaga* in immunocompetent patients with oral infection [6], as well as sepsis with vertebral osteomyelitis in immunosuppressed patients [4, 7]. Our patient was immunocompetent, maintained good oral hygiene, remained afebrile without infective symptoms, and experienced symptom onset more than six weeks after a minor dog bite, underscoring the indolent nature of the pathogen even in healthy hosts.

Capnocytophaga is a slow-growing organism that is difficult to cultivate and may evade conventional blood cultures [8]. Given this fastidious nature, PCR—particularly 16S rRNA sequencing—is a superior diagnostic tool [9]. In this case, both tissue cultures were negative, and only 16S rRNA testing confirmed the pathogen.

Management of *Capnocytophaga* infections depends on the site, severity, and patient comorbidities. Effective antimicrobial options include β -lactams/ β -lactamase inhibitor combinations, tetracyclines, clindamycin, fluoroquinolones, penicillins, linezolid, and rifampicin [5, 10]. Although treatment duration is not clearly defined, antimicrobial therapy of at least 6–8 weeks is generally recommended for bone and joint infections [5]. Our patient was initially treated with ciprofloxacin but switched to Augmentin Duo Forte due to adverse effects, ultimately achieving recovery after a 12-week course.

This case highlights that *Capnocytophaga* can cause indolent yet clinically significant infections in immunocompetent patients, underscoring the importance of obtaining a thorough exposure history in relevant clinical contexts. Moreover, when infection is strongly suspected and conventional testing is negative, 16S rRNA testing should be considered to establish the diagnosis and guide targeted antimicrobial therapy.

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Conflict of interest

Authors report no conflict of interest of any sort.

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Ethical approval

Not applicable.

Consent

Written informed consent obtained from the patient.

Guarantor

Nuwan Jasenthur Kankanamage.

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